Missing Light in Detached Eclipsing Binaries: the Case of KIC 2557430

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Abstract

A detached eclipsing binary's light curve and radial velocities give each component's radius and temperature, and so its luminosity, independently of its distance; with a Gaia parallax the system's apparent brightness follows, and a structure that intercepts part of the light would leave the system fainter than its components predict. We test the 117 systems of DEBCat with components below 8000 K within 1 kpc and renormalised unit weight error (RUWE) of 1.4 or less. The observed minus predicted G has a median of -0.04 mag and a robust scatter of 0.12 mag; 12 systems are too bright, by the light of third stars and blends, and one is too faint by 0.3 mag at 3σ: KIC 2557430, by 1.05 ± 0.19 mag. Gaia resolves it into the eclipsing pair, $G = 11.44$ at 343 pc, and a co-moving K star 2.4″ away at 339 pc, inside Kepler's pixels. The pair's colour and Gaia temperature, 6130 K with no extinction, lie 900 K below the published 7050 K primary, and Gaia's luminosities for the two sources sum to $2.9\,L_\odot$ against the solution's $9.7\,L_\odot$. The light is not missing from the system: the published temperatures, and the extinction the test inferred from them, are too high, and a spectroscopic orbit free of the companion's light decides the radii.

1 Introduction

A structure that intercepts part of a star's light and re-radiates it in the infrared leaves the star fainter in the optical than its size and temperature imply. For a single star the size comes from models or interferometry, but a detached eclipsing binary gives both components' radii from its light curve and its velocities, and their temperatures from spectra and the eclipse depths, with uncertainties of one or two per cent for the best systems [1]. With a Gaia parallax [2] the system's luminosity then predicts its apparent magnitude, and a shortfall is light that the system does not send us. Detached eclipsing binaries have been used the other way, to test Gaia's parallaxes against their own distances [3]; here the parallaxes are taken as given and the light is tested.

1.1 Contributions

1. A comparison of the predicted and observed Gaia G magnitudes of the 117 clean detached eclipsing binaries of DEBCat.
2. The resolution of the one system too faint, KIC 2557430, by Gaia's view of the system and its companion.

2 Data and method

DEBCat [1] lists 388 detached eclipsing binaries with masses and radii measured to a few per cent, and for 252 of them both components' luminosities and temperatures and a Gaia DR3 counterpart

Table 1: Gaia DR3 for the two sources of KIC 2557430: parallax, G, colour, RUWE, and the temperature, radius and luminosity of the astrophysical parameters (GSP-Phot and FLAME), which treat each source as a single star.

Source	ϖ (mas)	G	BP−RP	RUWE	$T_{\rm eff}$ (K)	R ($R_\odot$)	L ($L_\odot$)
2099133865420292992	2.916 ± 0.012	11.44	0.68	0.81	6131	1.31	2.20
2099133865420292480	2.946 ± 0.015	13.30	1.39	1.55	4698	1.31	0.74

through CDS XMatch on their SIMBAD names. For each component, $\log L$ and $\log T_{\rm eff}$ give an absolute G through bolometric corrections in temperature, and the two components summed and carried to the Gaia distance predict G. The extinction applied is the larger of Gaia's A_G and $2.7\,E(B-V)$, with $E(B-V)$ the catalogue's $B-V$ less the intrinsic colour of the components' combined light. We keep the 117 systems whose components are cooler than 8000 K, which lie within 1 kpc and whose RUWE is 1.4 or less, so that the parallax and the photometry belong to an unresolved, well-measured source; BG Ind, IM Per and the other systems of high RUWE are set aside. A system is too faint when $G_{\rm obs} - G_{\rm pred}$ exceeds 0.3 mag at 3σ, with the error from the luminosities, the parallax and the photometry.

3 Results

Across the 117 systems $G_{\rm obs} - G_{\rm pred}$ has a median of -0.04 mag and a robust scatter of 0.12 mag. Twelve systems are too bright, as light from a third star or a blend in Gaia's photometry makes them, and one is too faint: KIC 2557430 (KOI-6277), an F2 and F9 pair with a period of 1.298 d, by 1.05 ± 0.19 mag (5.6σ), a factor of 2.6 in flux.

Its published solution [4] gives radii of 1.88 and 1.23 $R_\odot$ and temperatures of 7050 and 6050 K, luminosities of 7.9 and 1.8 $L_\odot$, and the catalogue's $B-V$ of 0.58 against the 0.38 of those components set the extinction in the test at 0.54 mag; without it the system would be 1.6 mag too faint. Table 1 gives Gaia's view. The system is the visual double WDS J19074+3749AB, and Gaia resolves it into two sources at the same parallax: the eclipsing pair, $G = 11.44$, $G_{\rm BP} - G_{\rm RP} = 0.68$, 2.916 ± 0.012 mas, with RUWE 0.81 and a radial velocity that varies by 20 km s^{-1} over 20 transits, and a K star 2.4″ away, $G = 13.30$, $G_{\rm BP} - G_{\rm RP} = 1.39$, 2.946 ± 0.015 mas. The test matched the eclipsing pair's source; adding the K star brings the system to $G = 11.26$, still 0.86 mag below the prediction.

Gaia's colours tell against the published temperatures rather than for missing light. Gaia gives the eclipsing pair $A_G = 0.001$ mag, so the catalogue's $B-V$ of 0.58 is the pair's own colour, that of a star near 6000 K, where a 7050 K primary carrying four fifths of the light would give about 0.4; Gaia's photometric temperature for the pair is 6130 K. At its parallax the pair's light corresponds to 2.2 $L_\odot$, and the K star's to 0.7 $L_\odot$, together 2.9 $L_\odot$ against the solution's 9.7 $L_\odot$. A primary at 6100 K with the published radius would hold 4.4 $L_\odot$, which halves the deficit, and the rest lies within what radii from a spectroscopic orbit contaminated by a third star can carry, since the K star, 1.9 mag fainter in G, falls inside Kepler's 4″ pixels and any spectrograph's fibre or slit at that separation. An earlier analysis of the Kepler light curve read the system as a triple holding a γ Doradus pulsator and two flaring components [5], and the K star is a flaring component that such a reading needs.

4 Related work

Detached eclipsing binaries serve as distance standards, and their luminosities have been used to test Gaia's parallaxes [3]; searches for intercepted light around single stars use their infrared excess against models of their photospheres [6]. We find no earlier search of eclipsing binaries for light that their measured components should send and do not.

5 Conclusion

No clean detached eclipsing binary of DEBCat is fainter than its components predict beyond the scatter of the comparison, 0.12 mag, save KIC 2557430, and Gaia shows that its deficit belongs to its published solution: the pair's colour and temperature are those of a 6100 K star with no extinction, and a resolved co-moving K star sits inside the apertures that gave the solution. The test bounds light intercepted from the other 116 systems at about 0.36 mag, three times the scatter, a third of the flux. A new spectroscopic orbit of KIC 2557430 that separates the K star's light, with temperatures from high-resolution spectra, would place it among them.

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